Reading Regression Output

A short guide for people who did not take statistics.

What the residuals tell you

When the spread of the residuals grows with the fitted values, the model shows heteroscedasticity. Ordinary least squares still gives unbiased estimates, but the standard errors are wrong, so the confidence intervals and p-values cannot be trusted as printed.

A residual plot is the quickest check. Look for a funnel shape rather than an even band around zero.

Fixing it

Robust standard errors are the usual first step. They change the uncertainty around each coefficient, not the coefficients themselves.

When to worry

Mild unevenness rarely matters for a large sample. It matters most when decisions depend on whether one effect is significant at 5 percent.

Report what you checked and how, so a reader can judge for themselves.